

Linear Algebra and Calculus Lecture 13

Eric Sandin Vidal

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Eigenvalues and Eigenvectors

Let

$$A = \begin{bmatrix} 3 & -2 \\ 1 & 0 \end{bmatrix}, \quad u = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad \text{and} \quad v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Try to draw the images of u and v under multiplication by A . In fact, Av is just $2v$. So A only “stretches” or dilates v .

Definition. An **eigenvector** of an $n \times n$ matrix A is a nonzero vector x such that $Ax = \lambda x$ for some scalar λ . A scalar λ is called an **eigenvalue** of A if there is a nontrivial solution x of $Ax = \lambda x$; such an x is called an eigenvector corresponding to λ .

Example. Show that 7 is an eigenvalue of matrix $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$, and find the corresponding eigenvectors.

The scalar 7 is an eigenvalue of A if and only if the equation

$$Ax = 7x$$

has a nontrivial solution. This equation is equivalent to $Ax - 7x = 0$, or

$$(A - 7I)x = 0$$

To solve this homogeneous equation, form the matrix

$$A - 7I = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} - \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} -6 & 6 \\ 5 & -5 \end{bmatrix}$$

The columns of $A - 7I$ are obviously linearly dependent, so the equation has nontrivial solutions, confirming that 7 is an eigenvalue of A . To find the corresponding eigenvectors, perform row operations:

$$\begin{bmatrix} -6 & 6 & 0 \\ 5 & -5 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The general solution has the form $x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Each vector of this form with $x_2 \neq 0$ is an eigenvector corresponding to $\lambda = 7$.

The set of all solutions of

$$(A - \lambda I)x = 0$$

is just the null space of the matrix $A - \lambda I$. Thus, this set is a subspace of $\mathbb{R}^n$ and is called the **eigenspace** of A corresponding to λ . The eigenspace consists of the zero vector and all the eigenvectors corresponding to λ .

Notice that this doesn't tell us anything of how to find the eigenvalues, only the eigenvectors corresponding to an eigenvalue.

Theorem. *The eigenvalues of a triangular matrix are the entries on its main diagonal.*

Theorem. *If $v_1, \dots, v_r$ are eigenvectors that correspond to distinct eigenvalues $\lambda_1, \dots, \lambda_r$ of an $n \times n$ matrix A , then the set $\{v_1, \dots, v_r\}$ is linearly independent.*

Example. Find the eigenvalues of $A = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix}$. We must find all scalars λ such that the matrix equation

$$(A - \lambda I)x = 0$$

has a nontrivial solution. By the Invertible Matrix Theorem, this problem is equivalent to finding all λ such that the matrix $A - \lambda I$ is not invertible, where

$$A - \lambda I = \begin{bmatrix} 2 & 3 \\ 3 & -6 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 2 - \lambda & 3 \\ 3 & -6 - \lambda \end{bmatrix}$$

We know that this matrix fails to be invertible precisely when its determinant is zero. So the eigenvalues of A are the solutions of the equation

$$\det(A - \lambda I) = \det \begin{bmatrix} 2 - \lambda & 3 \\ 3 & -6 - \lambda \end{bmatrix} = 0$$

Thus,

$$\begin{aligned} \det(A - \lambda I) &= (2 - \lambda)(-6 - \lambda) - (3)(3) \\ &= -12 + 6\lambda - 2\lambda + \lambda^2 - 9 \\ &= \lambda^2 + 4\lambda - 21 \\ &= (\lambda - 3)(\lambda + 7) \end{aligned}$$

If $\det(A - \lambda I) = 0$, then $\lambda = 3$ or $\lambda = -7$. So the eigenvalues of A are 3 and -7.

Theorem. *(The Invertible Matrix Theorem DLC 2) A square matrix A is invertible if and only if*

(r) *The number 0 is not an eigenvalue of A .*

Diagonalization

The scalar equation $\det(A - \lambda I) = 0$ is called the **characteristic equation**. Since $\det(A - \lambda I)$ is a polynomial in λ , we call $\det(A - \lambda I)$ the **characteristic polynomial**.

Definition. If A and B are $n \times n$ matrices, then A is **similar** to B if there is an invertible matrix P such that

$$P^{-1}AP = B,$$

or, equivalently,

$$A = PBP^{-1}.$$

Writing Q for P^{-1} , we have

$$Q^{-1}BQ = A.$$

Thus, B is also similar to A , and we say simply that A and B are similar. Changing A into $P^{-1}AP$ is called a **similarity transformation**.

Theorem. If $n \times n$ matrices A and B are similar, then they have the same characteristic polynomial and hence the same eigenvalues (with the same multiplicities).

Definition. A square matrix A is said to be **diagonalizable** if A is similar to a diagonal matrix, that is, if $A = PDP^{-1}$ for some invertible matrix P and some diagonal matrix D .

Theorem. An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors. In fact, $A = PDP^{-1}$, with D a diagonal matrix, if and only if the columns of P are n linearly independent eigenvectors of A . In this case, the diagonal entries of D are eigenvalues of A that correspond, respectively, to the eigenvectors in P .

Theorem. If an $n \times n$ matrix has n distinct eigenvalues, then it is diagonalizable.

Remark. NOT THE OTHER WAY AROUND!

Theorem. Let A be an $n \times n$ matrix whose distinct eigenvalues are $\lambda_1, \dots, \lambda_p$.

1. For $1 \leq k \leq p$, the dimension of the eigenspace for λ_k is less than or equal to the multiplicity of the eigenvalue λ_k .
2. The matrix A is diagonalizable if and only if the sum of the dimensions of the eigenspaces equals n , and this happens if and only if (i) the characteristic polynomial factors completely into linear factors and (ii) the dimension of the eigenspace for each λ_k equals the multiplicity of λ_k .
3. If A is diagonalizable and B_k is a basis for the eigenspace corresponding to λ_k for each k , then the total collection of vectors in the sets $\mathcal{B}_1, \dots, \mathcal{B}_p$ forms an eigenvector basis for $\mathbb{R}^n$.

Example. Diagonalize the following matrix, if possible.

$$A = \begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 1 & 4 & -3 & 0 \\ -1 & -2 & 0 & -3 \end{bmatrix}$$

Since A is a triangular matrix, the eigenvalues are 5 and -3 , each with multiplicity 2. Using the method for finding null spaces, we find a basis for each eigenspace.

Basis for $\lambda = 5$:

$$v_1 = \begin{bmatrix} -8 \\ 4 \\ 1 \\ 0 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -16 \\ 4 \\ 0 \\ 1 \end{bmatrix}$$

Basis for $\lambda = -3$:

$$v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad v_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

The set $\{v_1, v_2, v_3, v_4\}$ is linearly independent, by the previous theorem. So the matrix

$$P = [v_1 \quad v_2 \quad v_3 \quad v_4]$$

is invertible, and $A = PDP^{-1}$ where

$$P = \begin{bmatrix} -8 & -16 & 0 & 0 \\ 4 & 4 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \quad D = \begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & -3 \end{bmatrix}$$